

Bhavya Modi

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EDUCATION

University of Waterloo

Waterloo, Canada | Aug. 2026

Bachelor of Computer Science (BCS)

Bachelor of Business Administration (BBA), Finance Concentration

TECHNICAL SKILLS

Languages: C/C++, Python, Rust, TypeScript, Java, SQL, Elixir

Systems & Infrastructure: Linux, Kafka, Redis, MongoDB, Grafana, PostgreSQL, Docker

Protocols & Frameworks: gRPC/Protobuf, FlatBuffers, WebSockets, GraphQL, React, Node.js, Tailwind

EXPERIENCE

Figure AI - Software Engineer Intern

San Jose, California | Sept. 2025 - Dec. 2025

- Engineered a **Linux** packet sniffer in **C++/libpcap** to decode live **FlatBuffer** telemetry at **8 kHz** with **zero dropped frames**, utilizing a pre-allocated ring buffer to save engineers **20 minutes per run**
- Overhauled robot **hardware-in-the-loop (HIL)** automated testing, cutting false-positive CI failures by **85%** with a **Python** validation service leveraging Kalman filters, moving averages, and dynamic thresholds
- Architected a unified **Grafana** observability pipeline across **5 data sources**, eliminating **2+ hours** of manual log correlation for **80+ engineers** triaging failures across **300 active robots**
- Optimized a **TypeScript** test-management platform, reducing test-publishing latency from **10 minutes to 50ms** by implementing database partitioning, compound indexing, and a **Redis** caching layer

theScore - Software Engineer Intern

Toronto, Canada | Jan. 2025 - May 2025

- Shipped **25** self-serve **GraphQL** endpoints for analytics access, cutting support tickets by **30/month** and eliminating backend data bottlenecks for downstream consumers
- Migrated high-traffic **Elixir** microservices to **gRPC** and **Protobuf**, reducing p95 inter-service latency from **75ms to 30ms** through binary serialization and typed service contracts
- Designed and published a reusable **React** and **Tailwind** component library adopted by **70 engineers** across **30+ pages**, standardizing UI design and accelerating feature delivery by **3 days per release**

DayForce - Software Developer Intern

Toronto, Canada | Jan. 2024 - May 2024

- Scaled the core data processing system, increasing transaction capacity by **40x** by replacing a legacy batch pipeline with an event-driven **Kafka** streaming pipeline to process incoming client messages in real-time
- Engineered automated incident recovery APIs for **MongoDB**, introducing durable state tracking and retry policies that reduced manual recovery from **10+ hours to under 1 hour per week**

CMiC - Software Engineer Intern

Toronto, Canada | Jan. 2023 - May 2023

- Orchestrated end-to-end **automated regression** testing, increasing test coverage to **95%** and cutting integration errors by **15%** through a nightly **JUnit** pipeline that caught failures before production
- Built **15 Java and SQL** reporting services and dashboards, giving stakeholders self-serve KPI access and saving the team **5+ hours per week** by replacing manual reporting requests with standardized views

PROJECTS

Warden - Robot Fleet Coordination

Demo | Aug. 2026

- Engineered a distributed fleet simulator for **1000** warehouse robots in **Rust** and **Python**, resolving network bottlenecks from continuous broadcasts by replacing centralized polling with localized routing decisions
- Reduced inter-robot network traffic by **85%** and guaranteed **zero collisions** across **100,000+** moves by querying a **2KB Ribbon filter** for occupancy and checking reserved cells against a **ring buffer**
- Built a real-time dashboard streaming CPU utilization, network queues, and fleet telemetry over **WebSockets**, enabling live performance monitoring and bottleneck diagnosis across **1000** active robots